

Varun Mukherjee

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Professional Summary

Software Engineer specializing in scalable backend architecture and AI-driven product development. Engineered resilient FinTech solutions, including prompt-based transaction systems, large-scale notification frameworks, and comprehensive auditing infrastructure. Resolved complex backend challenges by integrating emerging technologies and optimizing system performance for high-traffic environments.

Experience

Motive

Jul 2024 – Present

Software Engineer

Bangalore, India

- Directed the engineering of a prompt based transaction system, eliminating mobile app unlock friction in low-network areas and scaling to **25K+ active cards** and **100K+ transactions** in 4 months.
- Architected scalable notification services (WhatsApp, SMS) and an XXL-scale transaction alert framework, processing **500K+ alerts** monthly with customizable event triggers.
- Pioneered an **AI Development Life Cycle (AIDLC)** to augment traditional SDLC and developed **internal AI tools** for automated PR reviews utilizing agentic workflows, which increased developer productivity by **50%**
- Spearheaded a legacy roles and permissions overhaul (securing **200+ endpoints**) and deployed an org-wide auditing framework, reducing incident debugging time by **40%**.
- Orchestrated a cross-functional policy migration abstracting **30+ complex settings** into **4 core policies**, while mentoring interns and facilitating rigorous peer reviews.

Motive

Jan 2024 – Jul 2024

Software Engineer Intern

Bangalore, India

- Executed the zero-downtime migration of asynchronous background jobs to serverless AWS SQS via Shoryuken, stabilizing **50+ core queues**.
- Engineered a 'Missed Savings' tracker and integrated GasBuddy data mapping **150K+ stops**, increasing driver app savings tool engagement by **25%**.
- Implemented granular pre-transaction overrides, eliminating rigid company-wide restrictions and reducing related customer support tickets by **20%**.

Polymath AI

Jul 2023 – Jan 2024

Full Stack Developer

Remote

- Designed the architecture for 4 distinct AI platforms using Next.js and FastAPI, integrating **OpenAI APIs, RAG, LangChain, Vector Databases**, and advanced **prompt engineering**.
- Constructed robust solutions that successfully supported an active monthly user base exceeding **20K users**.

Education

Indian Institute of Information Technology (IIIT), Sri City

Dec 2020 – Jun 2024

Bachelor of Science in Computer Science (CGPA: 9.22)

Sri City, Andhra Pradesh

Technical Skills

Language: Java, Ruby, Python, C/C++, C#, Go, Javascript, Typescript, SQL, HTML/CSS

Frameworks: Ruby on Rails, FastAPI, Node.js, Express.js, React.js, Angular.js, Next.js, Redux, .Net, Full-stack development, Langchain, LangGraph, Kafka, Rspec, Minitest

Database: MySQL, PostgreSQL, MongoDB, NoSQL, AWS DynamoDB, PineconeDB, Redis

AI and tooling: RAG, Prompt Engineering, OpenAI APIs, Agentic AI, LLM, Agentic Skill Development

Software development: Modular monolith, Domain driven design, OOPs, Distributed systems, Event driven systems, Test driven development, Amazon SQS, Kafka, AWS, Docker, Git

Publications

IEEE Conference on Advanced Robotics and Its Social Impacts (ARSO)

May 2024

Lead Author: 3DHoNR - 3D Object Recognition using a Low-Dimensional Descriptor

IIIT Sri City